

Deriving the Oseen-Lamb Vortex: Three Methods

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Contents

1 The First Method: Finding $g(r, t)$ And Using A Similarity Variable	1
2 The Second Method: Using The Vorticity Transport Equation And The Frobenius Method	4
3 The Third Method: Green's Theorem And The Velocity Parcel in $\mathbb{R}^2$	7
References	9

Abstract

In great depth, we investigate three methods to derive the axially symmetric azimuthal velocity model: the Oseen-Lamb vortex. The first method is finding a velocity solution of the form, $u_\theta = \frac{\Gamma_0}{2\pi r} g(r, t)$ for some function, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. The second method is to solve the vorticity transport equation in polar coordinates, which yields a Sturm-Liouville differential equation with a lengthy process. The third method is by no means as robust as the first two, but it utilizes Green's theorem and a velocity parcel to obtain a separable DE in terms of circulation.

Introduction & Boundary Conditions:

The solution to the advectionless and pressureless Navier-Stokes equation given a unidirectional velocity field, $\vec{u} = \langle 0, u_\theta, 0 \rangle$, is the Lamb-Oseen vortex.

$$u_\theta(r, t) = \frac{\Gamma_0}{2\pi r} \left(1 - e^{-\frac{r^2}{4\nu t}} \right) \quad , \quad \Gamma_0, \nu \in \mathbb{R}^+ \quad (1)$$

with the boundary conditions,

$$\left\{ \begin{array}{ll} u(0, t) = 0 \quad , & u(r \rightarrow \infty, t) = 0 \\ u(r, 0) = \frac{\Gamma_0}{2\pi r} \quad , & u(r, t \rightarrow \infty) = 0 \end{array} \right\}$$

1 The First Method: Finding $g(r, t)$ And Using A Similarity Variable

Consider an azimuthal velocity solution of the form,

$$u_\theta(r, t) = \frac{\Gamma_0}{2\pi r} g(r, t) \quad (2)$$

and substitute (2) into the laminarized momentum equation,

$$\begin{aligned}
\frac{\partial u_\theta}{\partial t} &= \nu \left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r^2} \right) \\
\frac{\partial}{\partial t} \left(\frac{\Gamma_0}{2\pi r} g(r, t) \right) &= \nu \left(\frac{\partial^2}{\partial r^2} \left(\frac{\Gamma_0}{2\pi r} g(r, t) \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{\Gamma_0}{2\pi r} g(r, t) \right) - \frac{1}{r^2} \frac{\Gamma_0}{2\pi r} g(r, t) \right) \\
\frac{\Gamma_0}{2\pi r} \frac{\partial g}{\partial t} &= \nu \left(\left(2 \frac{\Gamma_0}{2\pi r^3} g - \frac{\Gamma_0}{2\pi r^2} \frac{\partial g}{\partial r} - \frac{\Gamma_0}{2\pi r^2} \frac{\partial g}{\partial r} + \frac{\Gamma_0}{2\pi r} \frac{\partial^2 g}{\partial r^2} \right) + \frac{1}{r} \left(-\frac{\Gamma_0}{2\pi r^2} g + \frac{\Gamma_0}{2\pi r} \frac{\partial g}{\partial r} \right) - \frac{1}{r^2} \frac{\Gamma_0}{2\pi r} g \right) \\
\frac{\partial g}{\partial t} &= \nu \left(\left(\frac{2}{r^2} g - \frac{2}{r} \frac{\partial g}{\partial r} + \frac{\partial^2 g}{\partial r^2} \right) + \frac{1}{r} \left(-\frac{1}{r} g + \frac{\partial g}{\partial r} \right) - \frac{1}{r^2} g \right) \\
\frac{\partial g}{\partial t} &= \nu \left(\frac{\partial^2 g}{\partial r^2} - \frac{1}{r} \frac{\partial g}{\partial r} \right)
\end{aligned} \tag{3}$$

We can invoke a similarity solution, $g(r, t) = f(\eta)$ with $\eta = \frac{r}{\sqrt{\nu t}}$, and substitute to obtain the following derivatives:

$$\begin{aligned}
\frac{\partial g}{\partial t} &= \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial t} = -f' \frac{rt^{-3/2}}{2\sqrt{\nu}} = -f' \frac{\eta}{2t} \\
\frac{\partial g}{\partial r} &= \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial r} = f' \frac{1}{\sqrt{\nu t}} \\
\frac{\partial^2 g}{\partial r^2} &= \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial r} \right) \\
&= \frac{\partial f}{\partial \eta} \underbrace{\frac{\partial}{\partial r} \left(\frac{\partial \eta}{\partial r} \right)}_{=0} + \frac{\partial \eta}{\partial r} \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial \eta} \right) \\
&= \frac{\partial \eta}{\partial r} \cdot \frac{\partial^2 f}{\partial \eta^2} \frac{\partial \eta}{\partial r} \\
&= f'' \frac{1}{\nu t}
\end{aligned} \tag{4}$$

The following separable ordinary differential equation is achieved by the substitution of the derivatives in (4) into (3),

$$\begin{aligned}
-f' \frac{\eta}{2t} &= \nu \left(\frac{f''}{\nu t} - \frac{f'}{r\sqrt{\nu t}} \right) \\
f'' + \left(\frac{\eta}{2} - \frac{1}{\eta} \right) f' &= 0
\end{aligned}$$

with the solution,

$$\begin{aligned}
\frac{d^2 f}{d\eta^2} &= -\left(\frac{\eta}{2} - \frac{1}{\eta}\right) \frac{df}{d\eta} \quad , \quad u(\eta) = f'(\eta) \\
\frac{du}{d\eta} &= -\left(\frac{\eta}{2} - \frac{1}{\eta}\right) u \\
\int \frac{du}{u} &= -\int \left(\frac{\eta}{2} - \frac{1}{\eta}\right) d\eta \\
\ln(u) &= -\left(\frac{\eta^2}{4} - \ln(\eta)\right) + C \\
u(\eta) &= C_1 e^{-\frac{\eta^2}{4} + \ln(\eta)} \\
f'(\eta) &= C_1 \eta e^{-\frac{\eta^2}{4}} \\
f(\eta) &= C_1 \int \eta e^{-\frac{\eta^2}{4}} d\eta \quad , \quad p = -\frac{\eta^2}{4} \quad , \quad dp = -\frac{\eta}{2} d\eta \\
f(\eta) &= C_1 \int \eta e^p \left(-\frac{2}{\eta}\right) dp \\
f(\eta) &= A e^{-\frac{\eta^2}{4}} + B \quad , \quad A = -2C_1 \\
\therefore \omega_{\hat{z}}(r, t) &= A e^{-\frac{r^2}{4\nu t}} + B
\end{aligned}$$

The constants, A and B , can be found by the circulation formula, defined as the vorticity flux integral over a circular area differential.

$$\begin{aligned}
\Gamma &= \int_A \vec{\omega}_{\hat{z}} \cdot dA = \int_0^\infty \omega_{\hat{z}} \cdot 2\pi r dr \\
&= 2\pi \int_0^\infty \left(A e^{-\frac{r^2}{4\nu t}} + B\right) r dr \\
&= 2\pi \left(\int_0^\infty A r e^{-\frac{r^2}{4\nu t}} dr + \int_0^\infty B r dr \right) \\
&= 2\pi \left(\int_0^\infty A r \left(\frac{2\nu t}{r}\right) e^{-u} du + B \frac{r^2}{2} \Big|_0^\infty \right) \\
&= 4\pi\nu A t \int_0^\infty e^{-u} du + \pi B r^2 \Big|_0^\infty \\
&= 4\pi\nu A t + \pi B r^2 \Big|_0^\infty \quad , \quad B = 0 \\
\Gamma_0 &= 4\pi\nu A t \\
\therefore A &= \frac{\Gamma_0}{4\pi\nu t}
\end{aligned}$$

where $\Gamma_0 \in \mathbb{R}^+$ is the initial circulation and $B = 0$ since there are no regions in the fluid with infinite circulation. The vorticity becomes the Gaussian distribution,

$$\omega_{\hat{z}}(r, t) = \frac{\Gamma_0}{4\pi\nu t} e^{-\frac{r^2}{4\nu t}} \tag{5}$$

Because $\omega_{\hat{z}} = \nabla \times u_{\theta} = \left(\frac{\partial u_{\theta}}{\partial r} + \frac{u_{\theta}}{r} \right) \hat{z}$, we can use the integrating factor, $e^{\int \frac{1}{r} dr} = r$, to isolate u_{θ} and substitute Equation (5) as follows:

$$\begin{aligned}
u_{\theta}(r, t) &= \frac{1}{r} \int r \omega_{\hat{z}} dr \\
&= \frac{\Gamma_0}{4\pi\nu t} \frac{1}{r} \int r e^{-\frac{r^2}{4\nu t}} dr \\
&= \frac{\Gamma_0}{4\pi\nu t} \frac{1}{r} \int r e^u \left(-\frac{2\nu t}{r} \right) du \\
&= -\frac{\Gamma_0}{2\pi} \frac{1}{r} \int e^u du \\
&= -\frac{\Gamma_0}{2\pi r} e^{-\frac{r^2}{4\nu t}} + C
\end{aligned} \tag{6}$$

Using the boundary condition, $u_{\theta}(0, t) = 0$, $C = \frac{\Gamma_0}{2\pi r}$, the solution to Navier-Stokes is obtained.

$$u_{\theta}(r, t) = \frac{\Gamma_0}{2\pi r} \left(1 - e^{-\frac{r^2}{4\nu t}} \right)$$

2 The Second Method: Using The Vorticity Transport Equation And The Frobenius Method

Taking the curl of the momentum equation,

$$\frac{\partial u_{\theta}}{\partial t} = \nu \left(\frac{\partial^2 u_{\theta}}{\partial r^2} + \frac{1}{r} \frac{\partial u_{\theta}}{\partial r} - \frac{u_{\theta}}{r^2} \right)$$

we have the $\hat{z}$ directional vorticity transport equation in $\mathbb{R}^2$:

$$\hat{z} : \frac{\partial \omega}{\partial t} = \nu \left(\frac{\partial^2 \omega}{\partial r^2} + \frac{1}{r} \frac{\partial \omega}{\partial r} \right) \tag{7}$$

We invoke a new similarity solution, $\eta = \frac{r}{\sqrt{4\nu t}}$ for $\omega_{\hat{z}}(r, t) = \frac{1}{t} f(\eta)$, and find each derivative:

$$\begin{aligned}
\frac{\partial \omega}{\partial t} &= \frac{\partial}{\partial t} \left(\frac{f}{t} \right) = \frac{\partial}{\partial t} \left(\frac{1}{t} \right) + \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial t} = -\frac{1}{t^2} f - \frac{2\nu r}{t(4\nu t)^{3/2}} f' \\
\frac{\partial \omega}{\partial r} &= \frac{1}{t} \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial r} = f' \frac{1}{t\sqrt{4\nu t}} \\
\frac{\partial^2 \omega}{\partial r^2} &= \frac{\partial}{\partial r} \left(\frac{1}{t} \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial r} \right) \\
&= \frac{1}{t} \frac{\partial f}{\partial \eta} \underbrace{\frac{\partial}{\partial r} \left(\frac{\partial \eta}{\partial r} \right)}_{=0} + \frac{1}{t} \frac{\partial \eta}{\partial r} \frac{\partial}{\partial r} \left(\frac{\partial f}{\partial \eta} \right) \\
&= \frac{1}{t} \frac{\partial \eta}{\partial r} \cdot \frac{\partial^2 f}{\partial \eta^2} \frac{\partial \eta}{\partial r} \\
&= f'' \frac{1}{4\nu t^2}
\end{aligned} \tag{8}$$

By substitution into (7), we have the ODE:

$$\begin{aligned}
-\frac{1}{t^2} f - \frac{2\nu r}{t(4\nu t)^{3/2}} f' &= \nu \left(f'' \frac{1}{4\nu t^2} + f' \frac{1}{r} \frac{1}{t\sqrt{4\nu t}} \right) \\
f'' + \left(\frac{1}{\eta} + 2\eta \right) f' + 4f &= 0
\end{aligned}$$

By inspection, we have a homogeneous Sturm-Liouville DE [2] of the form,

$$\frac{d}{dx} \left[p(x) \frac{d\phi}{dx} \right] + [q(x) + \lambda w(x)] \phi(x) = 0$$

with a general solution in the form of a linearly independent eigenfunction pair of solutions to the DE,

$$\phi(x) = c_1 \phi_1(x) + c_2 \phi_2(x)$$

To demonstrate with the integrating factor, $p(\eta) = e^{\int \frac{1}{\eta} + 2\eta d\eta} = \eta e^{\eta^2}$, we can get our DE in Sturm-Liouville form (where $q(x) = 0$),

$$\begin{aligned} \eta e^{\eta^2} f'' + \left(\frac{1}{\eta} + 2\eta \right) \eta e^{\eta^2} f' + 4\eta e^{\eta^2} f &= 0 \\ \frac{d}{d\eta} \left(\eta e^{\eta^2} f' \right) + 4\eta e^{\eta^2} f &= 0 \end{aligned}$$

and a general solution, $f(\eta) = c_1 f_1(\eta) + c_2 f_2(\eta)$. Using Frobenius's method [1], let our DE be,

$$y'' + \left(\frac{1}{x} + 2x \right) y' + 4y = 0$$

with the solution in the form, $y(x) = c_1 y_1(x) + c_2 y_2(x)$. Each derivative can be expressed as the series,

$$\begin{aligned} y(x) &= \sum_{k=0}^{\infty} c_k x^{k+r} \\ y'(x) &= \sum_{k=0}^{\infty} c_k (k+r) x^{k+r-1} \\ y''(x) &= \sum_{k=0}^{\infty} c_k (k+r)(k+r-1) x^{k+r-2} \end{aligned}$$

By substitution,

$$\begin{aligned} 0 &= \sum_{k=0}^{\infty} c_k (k+r)(k+r-1) x^{k+r-2} + \left(\frac{1}{x} + 2x \right) \sum_{k=0}^{\infty} c_k (k+r) x^{k+r-1} + 4 \sum_{k=0}^{\infty} c_k x^{k+r} \\ &= \sum_{k=0}^{\infty} c_k (k+r)(k+r-1) x^{k+r-2} + \left\{ \frac{1}{x} \sum_{k=0}^{\infty} c_k (k+r) x^{k+r-1} + 2x \sum_{k=0}^{\infty} c_k (k+r) x^{k+r-1} \right\} + 4 \sum_{k=0}^{\infty} c_k x^{k+r} \\ &= \sum_{k=0}^{\infty} c_k (k+r)(k+r-1) x^{k+r-2} + \sum_{k=0}^{\infty} c_k (k+r) x^{k+r-2} + 2 \sum_{k=0}^{\infty} c_k (k+r) x^{k+r} + 4 \sum_{k=0}^{\infty} c_k x^{k+r} \\ &= \sum_{k=0}^{\infty} c_k [(k+r)(k+r-1) + (k+r)] x^{k+r-2} + \sum_{k=0}^{\infty} 2c_k (k+r+2) x^{k+r} \\ &= \sum_{k=0}^{\infty} c_k (k+r)^2 x^{k+r-2} + \sum_{k=0}^{\infty} 2c_k (k+r+2) x^{k+r} \\ &= \sum_{k=0}^{\infty} c_k (k+r)^2 x^{k+r-2} + \sum_{k-2=0}^{\infty} 2c_{k-2} (k-2+r+2) x^{k-2+r} \\ &= \sum_{k=0}^{\infty} c_k (k+r)^2 x^{k+r-2} + \sum_{k=2}^{\infty} 2c_{k-2} (k+r) x^{k+r-2} \\ &= c_0 r^2 x^{r-2} + c_1 (1+r)^2 x^{r-1} + \sum_{k=2}^{\infty} [c_k (k+r)^2 + 2c_{k-2} (k+r)] x^{k+r-2} \end{aligned}$$

With $c_1 = 0$, we have a double root in which $r^2 = 0$ and $r_1 = r_2 = 0$, thus the indicial equation becomes,

$$c_k k^2 + 2c_{k-2}k = 0$$

$$\therefore c_k = -\frac{2c_{k-2}}{k}, \quad k \geq 2$$

Testing the first few values of c_k , we observe the pattern,

$$\begin{array}{lll} c_1 = 0 & c_4 = -\frac{2}{4}c_2 = \frac{1}{2}c_0 & c_7 = 0 \\ c_2 = -c_0 & c_5 = 0 & c_8 = -\frac{2}{8}c_6 = -\frac{1}{4}\left(-\frac{1}{3}\left(\frac{1}{2}c_0\right)\right) \\ c_3 = 0 & c_6 = -\frac{1}{3}c_4 = -\frac{1}{3}\left(-\frac{1}{2}c_2\right) = -\frac{1}{3}\left(\frac{1}{2}c_0\right) & c_9 = 0 \end{array}$$

Therefore, $\{-c_0, \frac{1}{2!}c_0, -\frac{1}{3!}c_0, \frac{1}{4!}c_0, -\frac{1}{5!}c_0, \dots\}$ and the first solution, $y_1(x)$, becomes,

$$\begin{aligned} y_1(x) &= |x|^{r_1} \sum_{k=0}^{\infty} c_k x^k, \quad c_{2m-1} = 0, \quad m \in \mathbb{N} \\ &= (1) \left(c_0 - \frac{c_0}{1!}x^2 + \frac{c_0}{2!}x^4 - \frac{c_0}{3!}x^6 + \dots \right) \\ &= \sum_{k=0}^{\infty} c_0 \frac{(-x^2)^k}{k!} \\ &= e^{-x^2} \text{ where, } c_0 = 1 \end{aligned}$$

The second solution, $y_2(x)$, can be found by substituting $y_2(x) = u(x)e^{-x^2}$ into the original DE:

$$\begin{aligned} 0 &= y'' + \left(\frac{1}{x} + 2x\right)y' + 4y \\ &= \left[u''e^{-x^2} - 4xu'e^{-x^2} - 2ue^{-x^2} + 4x^2ue^{-x^2} \right] + \left(\frac{1}{x} + 2x\right) \left[u'e^{-x^2} - 2xue^{-x^2} \right] + 4ue^{-x^2} \\ &= u'' - 4xu' - 2u + 4x^2u + \left(\frac{1}{x} + 2x\right)(u' - 2xu) + 4u \\ &= u'' - 4xu' - 2u + 4x^2u + \left(\frac{1}{x}u' - 2u + 2xu' - 4x^2u\right) + 4u \\ &= u'' + \frac{1}{x}u' - 2xu' \\ u'' &= -\left(\frac{1}{x} - 2x\right)u', \quad \text{solve with, } w(x) = u'(x) \\ u(x) &= \frac{C_1}{2}\text{Ei}(x^2), \quad C_2 = C_1/2 \\ \therefore y_2(x) &= C_2\text{Ei}(x^2)e^{-x^2} \end{aligned}$$

The general solution is then

$$\begin{aligned} y(x) &= c_1 y_1(x) + c_2 y_2(x) \\ y(x) &= c_1 e^{-x^2} + c_2 \text{Ei}(x^2)e^{-x^2} \quad \text{denoting, } y(x) = f(\eta), \quad x = \eta \\ \therefore f(\eta) &= c_1 e^{-\eta^2} + c_2 \text{Ei}(\eta^2)e^{-\eta^2} \end{aligned}$$

Because $\lim_{x \rightarrow 0} \text{Ei}(x^2)$ is not well defined, then $c_2 = 0$. Substituting η and $f(\eta)$ into $\omega_{\hat{z}}(r, t) = \frac{1}{t}f(\eta)$ yields,

$$\omega_{\hat{z}}(r, t) = c_1 e^{-\frac{r^2}{4\nu t}}$$

which leads to Equation (5) and the method in Equation (6).

3 The Third Method: Green's Theorem And The Velocity Parcel in $\mathbb{R}^2$

This method is speculative, superfluous, and by no means robust in comparison to the previous two methods explored.

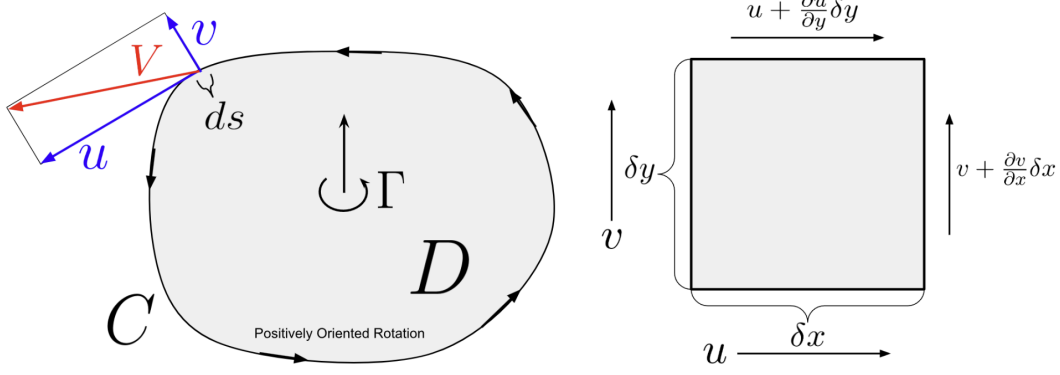


Figure 1: (Left) A general curve, C , enclosing a region, D , whose perimeter differential, ds , has x and y velocity components. (Right) A parcel of this region with those components plus the differentials.

For a velocity vector $\vec{V} = \langle u, v \rangle$ along the curve C , we find that by Green's theorem in Cartesian coordinates, the vorticity in the region D equals the flux of the vorticity with respect to the infinitesimal area.

$$\Gamma = \oint_C \vec{V} \cdot ds = \oint_C (u dx + v dy) = \iint_D \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy = \iint_D \|\vec{\omega}_z\| dA = \iint_D \vec{\omega}_z \cdot d\vec{A}$$

Taking differentials in terms of " δ ," we find the variation in circulation to be,

$$\begin{aligned} \delta\Gamma &= - \left[u\delta x + \left(v + \frac{\partial v}{\partial x}\delta x \right) \delta y \right] + \left[v\delta y + \left(u + \frac{\partial u}{\partial y}\delta y \right) \delta x \right] \\ &= - \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \delta x \delta y \\ &= -\vec{\omega}_z \delta A \\ \frac{\delta\Gamma}{\delta A} &= -\vec{\omega}_z \end{aligned} \tag{9}$$

Suppose that $\Gamma(r, t) = \vec{\omega}_z(r, t)a(t)$ in Equation (9) with a time-dependent area function, $a(t) = \pi R(t)^2$, and vortex radius, $R(t)$. The derivative, $\frac{d\Gamma}{dA}$, implies that $A(r) \in \Gamma(r, t)$. Then,

$$\begin{aligned} \frac{\delta\Gamma}{\delta A} &= -\frac{\Gamma(r, t)}{a(t)} \\ \frac{d\Gamma}{dA} &= -\frac{\Gamma(r, t)}{a(t)} \\ \int \frac{d\Gamma}{\Gamma} &= - \int \frac{dA}{a(t)} \\ \Gamma(r, t) &= e^{-\frac{A(r)}{a(t)} + C} \\ \Gamma(r, t) &= C_1 e^{-\frac{A(r)}{a(t)}} \quad , \quad \Gamma(0, t) = \Gamma_0 = C_1 \\ \Gamma(r, t) &= \Gamma_0 e^{-\frac{A(r)}{a(t)}} \end{aligned}$$

Because $\Gamma(r, t) = \pi R(t)^2 \omega_z(r, t)$ and $\Gamma_0 = \pi R(t)^2 \omega_z(t)$,

$$\begin{aligned}\pi R(t)^2 \omega_z(r, t) &= \Gamma_0 e^{-\frac{r^2}{R(t)^2}} \\ \omega_z(r, t) &= \frac{\Gamma_0}{\pi R(t)^2} e^{-\frac{r^2}{R(t)^2}}\end{aligned}\tag{10}$$

We can use the vorticity transport equation to solve for $R(t)$, first by taking each derivative,

$$\begin{aligned}\frac{\partial \omega}{\partial t} &= \frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2r^2}{R^3} - \frac{2}{R} \right) \frac{dR}{dt} \\ \frac{1}{r} \frac{\partial \omega}{\partial r} &= -\frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2}{R^2} \right) \\ \frac{\partial^2 \omega}{\partial r^2} &= -\frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2}{R^2} - \frac{4r^2}{R^4} \right)\end{aligned}$$

and second, by substituting into Equation (7). We find $R(t)$ by the separable DE:

$$\begin{aligned}\frac{\partial \omega}{\partial t} &= \nu \left(\frac{\partial^2 \omega}{\partial r^2} + \frac{1}{r} \frac{\partial \omega}{\partial r} \right) \\ \frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2r^2}{R^3} - \frac{2}{R} \right) \frac{dR}{dt} &= \nu \left[-\frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2}{R^2} - \frac{4r^2}{R^4} \right) - \frac{\Gamma_0}{\pi R^2} e^{-\frac{r^2}{R^2}} \left(\frac{2}{R^2} \right) \right] \\ \left(\frac{2r^2}{R^3} - \frac{2}{R} \right) \frac{dR}{dt} &= \nu \left[\frac{4r^2}{R^4} - \frac{2}{R^2} - \frac{2}{R^2} \right] \\ \left(\frac{2r^2}{R^3} - \frac{2}{R} \right) \frac{dR}{dt} &= \nu \frac{2}{R} \left(\frac{2r^2}{R^3} - \frac{2}{R} \right) \\ \frac{dR}{dt} &= \frac{2\nu}{R} \\ \int R dR &= 2\nu \int dt \\ \frac{R^2}{2} &= 2\nu t + C \\ \therefore R(t) &= \sqrt{4\nu t + R_0^2}\end{aligned}$$

It is not commonly found in physics journals that the initial vortex radius, R_0 , is used, though it has interesting implications when examining confined cyclonic flow. For the purpose of this paper, $R_0 = 0$ so that Equation (10) becomes Equation (5).

$$\omega_z(r, t) = \frac{\Gamma_0}{4\pi\nu t} e^{-\frac{r^2}{4\nu t}}$$

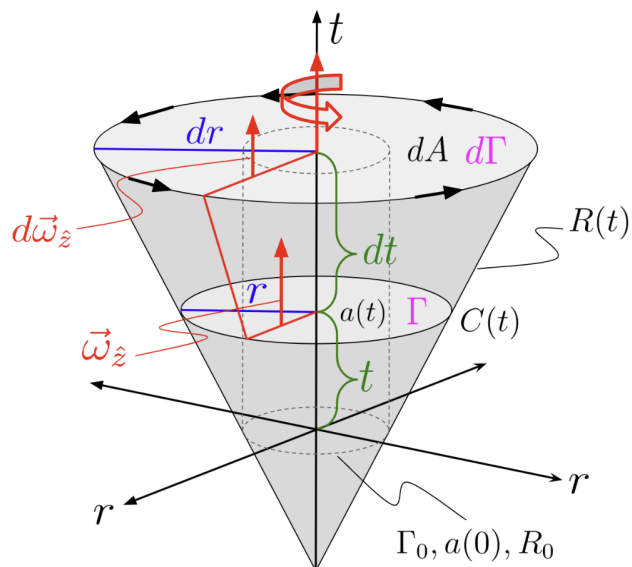


Figure 2: A curve, $C(t)$, enclosing the circular region at time, t , whose area and radius is $a(t)$ and $R(t)$.

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